

Informed principal and mechanism design with default continuation game

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Introduction

- Informed principal problem (Myerson 83), (Maskin Tirole 90, 92):
 - ▶ principal proposes a mechanism,
 - ▶ agent accepts, in which case the mechanism is implemented, or rejects,
 - ▶ typically, rejection leads to a fixed outside option.
- This paper: rejection leads to a default continuation game g :
 - ▶ entrepreneur seeking financing from a private investor before a public offering,
 - ▶ bidder proposing collusion before an auction,
 - ▶ duopolist proposing a cartel agreement,
 - ▶ bargaining with mechanisms, where rejection leads to another round of bargaining.

Introduction

Assumptions and goal

- Equilibrium outcomes of g depend on the information revealed at the proposal and at the acceptance stage.
 - ▶ The offer is a signal about the principal's type: beliefs $p_1 \rightarrow q_1$.
 - ▶ The accept-reject decision reveals information about the agent's type: beliefs $p_2 \rightarrow q_2^R$.
- Three main assumptions:
 - ▶ transferable utility,
 - ▶ principal's types can be verified,
 - ▶ mediator.
- Goal: the worst equilibrium payoff of the principal.
- Intermediate goal: solve the mechanism design problem: uninformed designer chooses a mechanism to maximize the worst equilibrium payoff of the principal, $v^*(p)$.

Introduction

Results

- **Theorem 1:** The worst equilibrium payoff of the informed principal is equal to $\text{cav}_1 v^*(p)$:
 - ▶ agent without private information,
 - ▶ agent with private information: mediator who names beliefs.
- **Theorem 2:** Characterization of $v^*(p)$
 - ▶ under Nondominance: $v^*(p)$ is the principal's optimal payoff subject to an IR constraint.
 - ▶ under "regularity": mediator + flexible IR constraint.

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 - ▶ under “regularity”: mediator + flexible IR constraint.

Introduction

Literature

- Informed principal:
 - ▶ (Myerson 83), (Maskin Tirole 90, 92), (Mylovanov Tröger 12, 14), (Dosis 22),
 - ▶ verifiable types: (Koessler Skreta 19, 23).
- Mechanisms competing with a default game:
 - ▶ ratifiability (Cramton Palfrey 95), (Celik Peters 11),
 - ▶ uninformative breakdown (Gerardi Myerson 07), informational punishment (Balzer Schneider 23),
 - ▶ mechanism design of peace (Fey Ramsay 11), (Hörner Morelli Squintani 15).
- Belief-based methods:
 - ▶ (Aumann Maschler 95), (Kamenica Gentzkow 11), (Lipnowski Ravid 20).
- Here: transferable utility, verifiable principal types, arbitrary default game g .

Outline

- 1 Introduction
- 2 **Model**
- 3 Informed principal
- 4 Mechanism design problem

Model

Environment

- Two players: principal (1, she) and agent (2, he), sometimes third player (“mediator”).
- Types $t_i \in T_i$, finite, independent; beliefs $p \in \Delta T = \Delta T_1 \times \Delta T_2$.
- Preferences: $u_i(x, t_i, t_{-i})$.
 - ▶ outcome $x \in X$ - closed convex subset of a Euclidean space.
- Allocation $\chi : T_1 \times T_2 \rightarrow X$; payoff vector $u \in \mathbb{R}^{T_1 \cup T_2}$:

$$u_i(t_i) = \sum_{t_{-i}} p_{-i}(t_{-i}) u_i(\chi(t_i, t_{-i}), t_i, t_{-i}).$$

- *Environment*: correspondence of feasible and IC payoffs

$$\mathcal{IC}(p) = \{u : u \text{ is the payoff vector of an allocation that is IC at } p\}.$$

- The geometry of the correspondence $\mathcal{IC}(\cdot)$ is the true “parameter” of the model.

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Model

Assumptions on environment

- *Transferable utility*: if $u \in \mathcal{IC}(p)$, then for each $\tau \in \mathbb{R}$,

$$(u_1 + \tau 1_1, u_2 - \tau 1_2) \in \mathcal{IC}(p).$$

- *Principal's verifiability*: if $u \in \mathcal{IC}(p)$, then for each $\tau \in \mathbb{R}^{T_1}$ st.
 $p_1 \cdot \tau = 0$,

$$(u_1 + \tau, u_2) \in \mathcal{IC}(p).$$

- ▶ utility can be transferred between the principal's types,
- ▶ no interim incentive constraints for the principal inside the mechanism,
- ▶ incentive constraints at the proposal stage and in the default game g are not affected.

Model

Mechanisms

- Mechanism $m = (M_1, M_2, M_M, x_m)$:
 - ▶ message sets for the players and for the mediator,
 - ▶ outcome function $x_m : M_1 \times M_2 \times M_M \rightarrow X$,
 - ▶ public randomization.
- Equilibrium payoff correspondence:

$$m : \Delta T \rightrightarrows \mathbb{R}^{T_1 \cup T_2}, \quad m(p) \subseteq \mathcal{IC}(p),$$

- ▶ u.h.c., non-empty valued, convex valued.
- Mediator: the only role is to name the beliefs.
 - ▶ If $\mu : \Delta T \rightarrow \mathcal{G}$ is st. $\mu(p|p)$ is u.h.c., then there is $m \in \mathcal{G}$ with $m(p) = \mu(p|p)$.

Model

Games

- Default continuation game g : defined as a mechanism, fixed.
- Accept-reject game $C_2(m, g)$: agent chooses between offer m and default g .
- Informed principal game:

$$I(g) = C_1(\{C_2(m, g) : m \text{ is a mechanism}\}).$$

- Perfect Bayesian Equilibrium:
 - ▶ players do not signal what they do not know,
 - ▶ mediator has the same beliefs as the players.
- Mechanism design problem:

$$v^*(p) = \sup_{m \in \mathcal{G}} \min_{u \in C(m, g)(p)} p_1 \cdot u_1.$$

- ▶ the designer does not choose the equilibrium of the accept-reject game,
- ▶ the designed mechanism does not reveal information about the principal.

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Theorem 1

Under the three main assumptions,

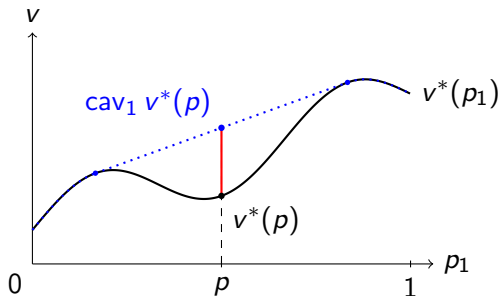
$$\inf_{u \in I(g)(p)} p_1 \cdot u_1 = \text{cav}_1 v^*(p; g).$$

If $T_2 = \{\}$, the mediator is not needed.*

- cav_1 is concavification along the principal's beliefs.

Informed principal

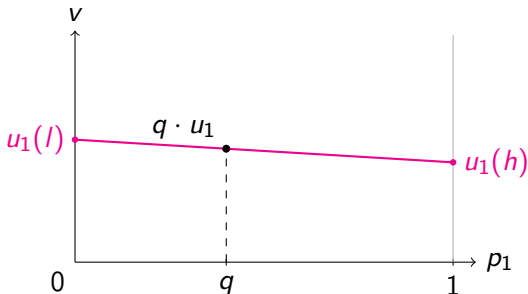
Comments



- The same form as Bayesian persuasion (Kamenica Gentzkow 11), but the argument is more similar to splitting lemma in repeated games (Aumann Maschler 95)
 - ▶ a mechanism that promises revelation and pays $cav_1 v^*(p) > v^*(p)$ to the principal may be rejected.
 - ▶ instead, (partial) revelation of information happens in equilibrium.
- Worst eq. payoff, but $cav_1 v^*(p) > v^*(p)$ is possible: compare with (Maskin Tirole 92).

Informed principal

Verifiability: choosing the slope

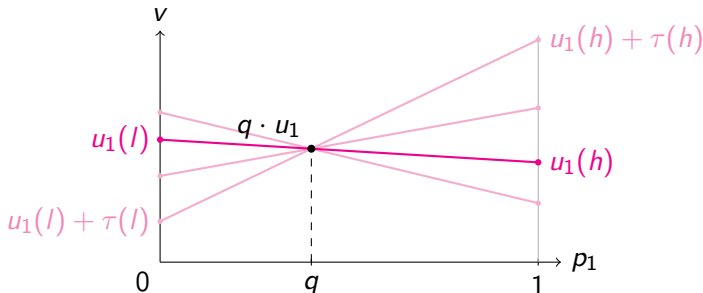


- Expected payoff $p_1 \mapsto p_1 \cdot u_1$ from u_1 .
- Verifiability: add transfers τ with $q \cdot \tau = 0$,

$$p_1 \cdot (u_1 + \tau) = p_1 \cdot u_1 + (p_1(h) - q(h))(\tau(h) - \tau(l)).$$

Informed principal

Verifiability: choosing the slope

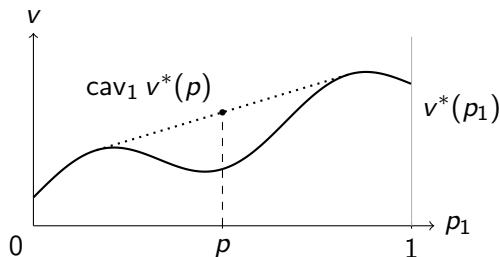


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Informed principal

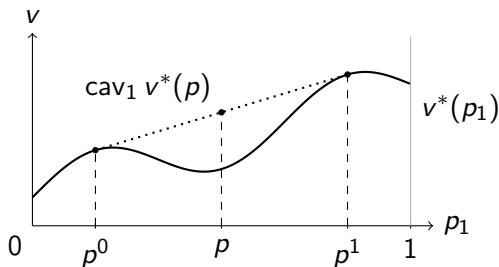
Proof: upper bound



- Upper bound: construct an equilibrium that attains $\text{cav}_1 v^*(p)$.
- Find beliefs p^k and weights α_k st. $p = \sum_k \alpha_k p^k$ and $\text{cav}_1 v^*(p) = \sum_k \alpha_k v^*(p^k)$.

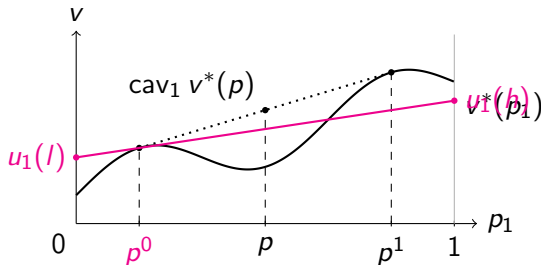
Informed principal

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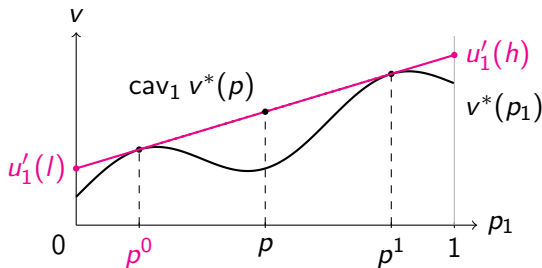


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- Find beliefs p^k and weights α_k st. $p = \sum_k \alpha_k p^k$ and $\text{cav}_1 v^*(p) = \sum_k \alpha_k v^*(p^k)$.
- Find mechanism m^0 with payoff vector u_1 such that

$$\min_{u \in C(m^0, g)(p^0)} p^0 \cdot u_1 = p^0 \cdot u_1 = v^*(p^0)$$

Informed principal

Proof: upper bound

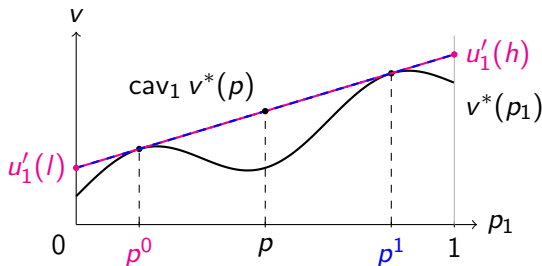


- Upper bound: construct an equilibrium that attains $\text{cav}_1 v^*(p)$.
- Find beliefs p^k and weights α_k st. $p = \sum_k \alpha_k p^k$ and $\text{cav}_1 v^*(p) = \sum_k \alpha_k v^*(p^k)$.
- Use verifiability to replace u by with payoff vector u'_1 tangent to $\text{cav}_1 v^*$,

$$\min_{u \in C(m^{0'}, g)(p^0)} p^0 \cdot u_1 = p^0 \cdot u'_1 = v^*(p^0)$$

Informed principal

Proof: upper bound

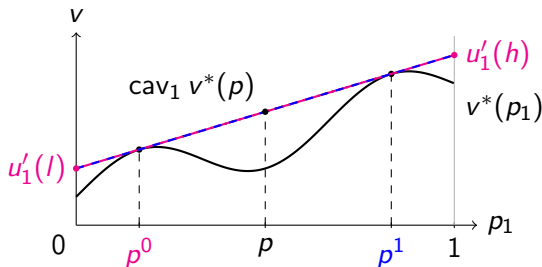


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- Find beliefs p^k and weights α_k st. $p = \sum_k \alpha_k p^k$ and $\text{cav}_1 v^*(p) = \sum_k \alpha_k v^*(p^k)$.
- The same for p^1 : mechanism $m^{1'}$ with the same payoff vector u'_1 ,

$$\min_{u \in C(m^{1'}, g)(p^1)} p^1 \cdot u_1 = p^1 \cdot u'_1 = v^*(p^1)$$

Informed principal

Proof: upper bound



- Upper bound: construct an equilibrium that attains $\text{cav}_1 v^*(p)$.
- Find beliefs p^k and weights α_k st. $p = \sum_k \alpha_k p^k$ and $\text{cav}_1 v^*(p) = \sum_k \alpha_k v^*(p^k)$.
- Mechanisms $m^{k'}$ with $(u'_1, u'_2) \in C(m^{k'}, g)(p^k)$: each type of the principal is indifferent.
- Equilibrium: principal randomizes, type t offers $m^{k'}$ with prob. $\alpha(k|t)$ st. posterior is p^k ; payoff $p \cdot u'_1 = \text{cav}_1 v^*(p)$.

Informed principal

Proof: upper bound

- Off-path offer m : choose the agent's posterior after m to deter the deviation.
- Let $c(q) = \{u_1 : u \in C(m, g)(q)\}$, $q \in \Delta T_1$. By definition of v^* and tangency of u'_1 ,

$$\min_{u \in c(q)} q \cdot u \leq v^*(q) \leq q \cdot u'_1.$$

Lemma (Gale-Nikaido-Debreu)

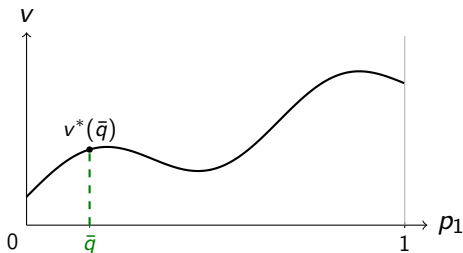
Suppose $c : \Delta T_1 \rightrightarrows \mathbb{R}^{T_1}$ is u.h.c., non-empty, convex and compact valued, and $\min_{u \in c(q)} q \cdot u \leq q \cdot l$ for each q . Then, there are p^ and $u \in c(p^*)$ st. $u \leq l$.*

- Apply with $l = u'_1$: assign posterior p^* to the offer m and continuation payoff $u \leq u'_1$.
- No type of the principal gains from the deviation.

Informed principal

Lower bound: menu $M(\bar{q})$

- Suppose the agent has no private information, $T_2 = \{*\}$.

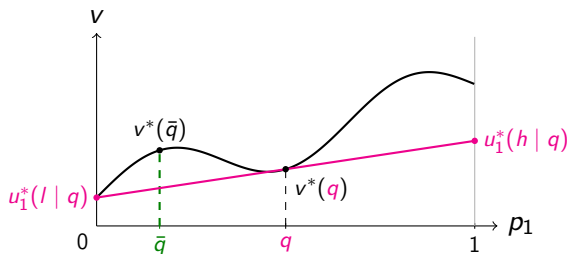


- Design belief $\bar{q}$: menu $M(\bar{q})$ that secures $v^*(\bar{q})$ in $\bar{q}$ -expectation.

Informed principal

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- Design belief $\bar{q}$: menu $M(\bar{q})$ that secures $v^*(\bar{q})$ in $\bar{q}$ -expectation.
- For each belief $q \in \Delta T_1$, let

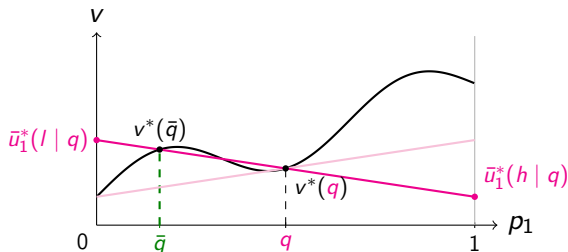
$$g_2^*(q) = \max\{u_2 : u \in g(q)\},$$

$$u^*(q) = \arg \max\{q \cdot u_1 : u \in \mathcal{IC}(q), u_2 \geq g_2^*(q)\}.$$

Informed principal

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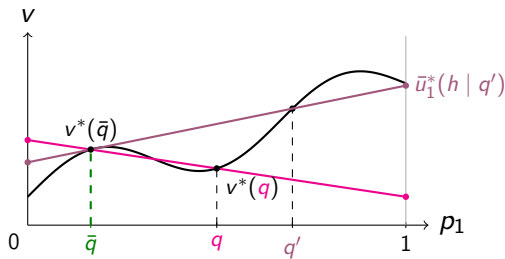
- Design belief $\bar{q}$: menu $M(\bar{q})$ that secures $v^*(\bar{q})$ in $\bar{q}$ -expectation.
- Use verifiability and transfers to find $\bar{u}^*(q)$ st.

$$\bar{u}_2^*(q) \geq g_2^*(q), \quad q \cdot \bar{u}_1^*(q) = v^*(q), \quad \bar{q} \cdot \bar{u}_1^*(q) \geq v^*(\bar{q}).$$

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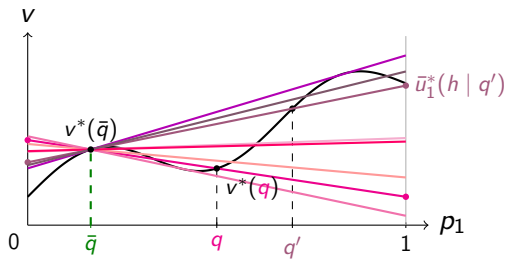


- Design belief $\bar{q}$: menu $M(\bar{q})$ that secures $v^*(\bar{q})$ in $\bar{q}$ -expectation.
- The same for each q' : all lines pass through $(\bar{q}, v^*(\bar{q}))$.

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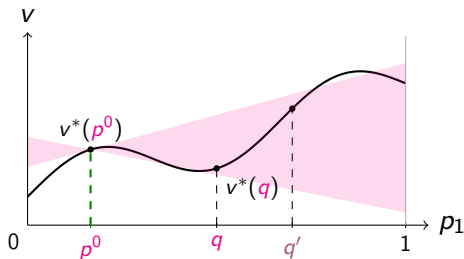


- Design belief $\bar{q}$: menu $M(\bar{q})$ that secures $v^*(\bar{q})$ in $\bar{q}$ -expectation.
- Menu $M(\bar{q}) = \{\bar{u}^*(q) : q \in \Delta T_1\}$:
 - for each belief q , an allocation that gives the agent more than $g_2^*(q)$
 $\implies$ will be accepted,
 - $\bar{q} \cdot \bar{u}_1^*(q) \geq v^*(\bar{q})$ for all allocations in the menu.

Informed principal

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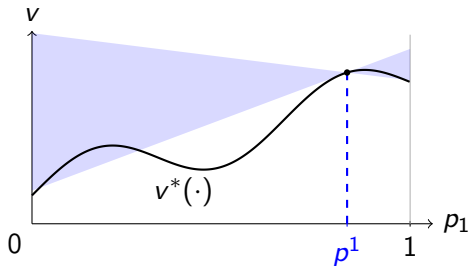
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- Design belief $\bar{q}$: menu $M(\bar{q})$ that secures $v^*(\bar{q})$ in $\bar{q}$ -expectation.
- $\bar{q} = p^0$: menu $M(p^0)$ secures $v^*(p^0)$ in p^0 -expectation.

Informed principal

Menus $M(p^0)$ and $M(p^1)$

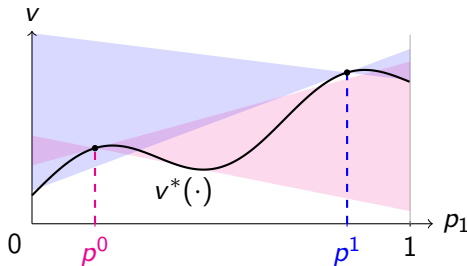


- The same for p^1 : menu $M(p^1)$ secures $v^*(p^1)$ in p^1 -expectation.
- Type t offers $M(p^k)$ with prob. $\alpha(k|t)$; posterior after $M(p^k)$ is p^k .
- For any beliefs and continuation equilibria, with $\alpha_k = \sum_t p(t)\alpha(k|t)$,

$$\sum_k \alpha_k p^k \cdot u_1^k \geq \sum_k \alpha_k v^*(p^k) = \text{cav}_1 v^*(p).$$

Informed principal

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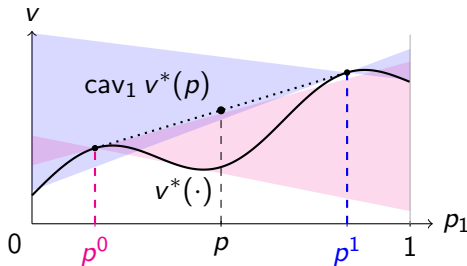


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Informed principal

Privately informed agent

- If $|T_2| \geq 2$, $v^*(q)$ may not be implementable by a single allocation.
- We use the mediator:

- In each case, the offer is accepted at true beliefs and the principal secures $v^*(\bar{q})$ at belief $\bar{q}$.

Informed principal

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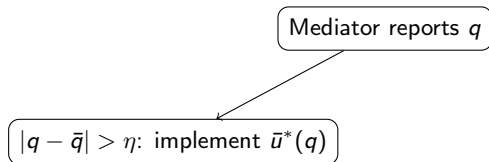
Mediator reports q

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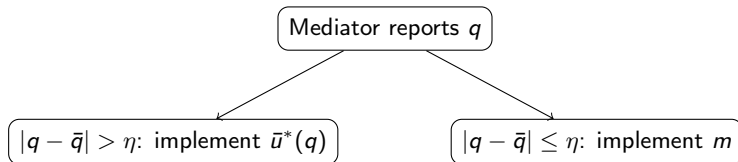
- always accepted: $u_2 > \max g_2$,
- transfers set the $\bar{q}$ -expected payoff:
 $\bar{q} \cdot \bar{u}_1^*(q) \geq v^*(\bar{q})$.

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m and $\eta > 0$ chosen so that

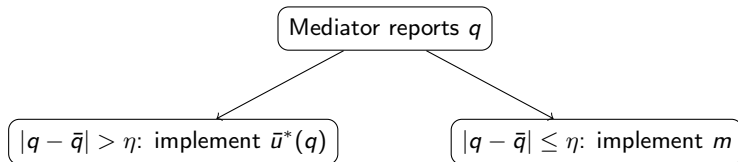
$$\min_{u \in C(m, g)(q)} \bar{q} \cdot u_1 \geq v^*(\bar{q}).$$

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Mechanism design

Problem

- Mechanism design problem:

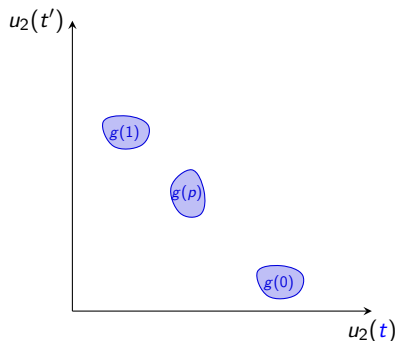
$$v^*(p) = \sup_{\text{mechanism } m \in \mathcal{G}} \min_{u \in C(m, g)(p)} p_1 \cdot u_1.$$

- Equilibria of the accept-reject game $C(m, g)$ depend only on the agent's payoffs u_2 in m and g .
- belief profile (p_1, p) , where p_1 fixed; agent's beliefs $p = p_2$.

Mechanism design

g payoff correspondence

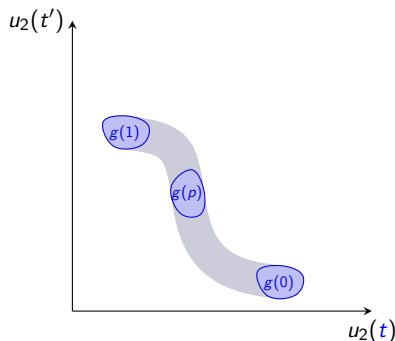
- Agent payoffs in $g(p)$,
 $p = p(t) = 1 - p(t')$.
- $B_g = \bigcup_q g(q)$ body of g
- F_g frontier of g
- enhanced payoffs
- flipped orientation: same curve,
reversed beliefs
 - ▶ cheap-talk equilibrium
 - ▶ public randomization equilibria
 - ▶ frontier of flipped g : a single point



Mechanism design

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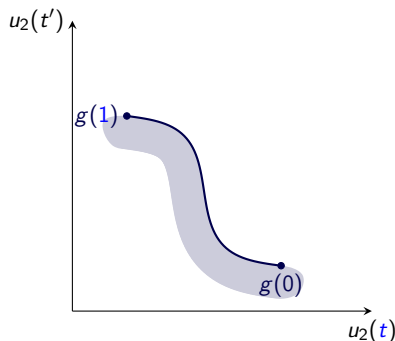
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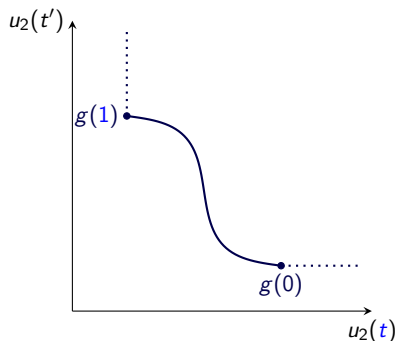
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Mechanism design

g payoff correspondence

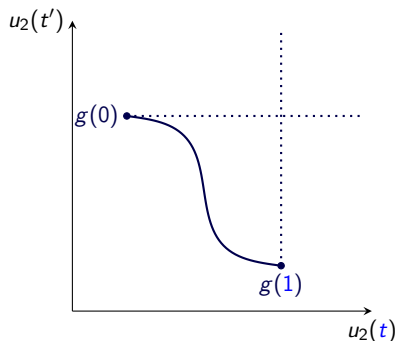
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- $B_g = \bigcup_q g(q)$ body of g
- F_g frontier of g
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 - cheap-talk equilibrium
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Mechanism design

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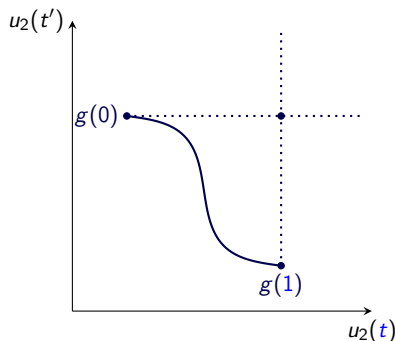
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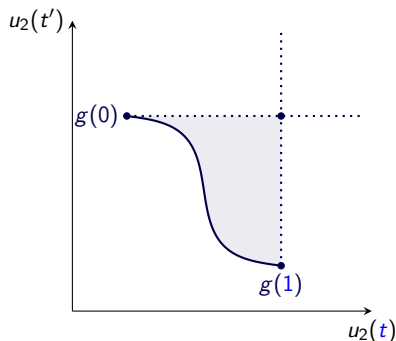
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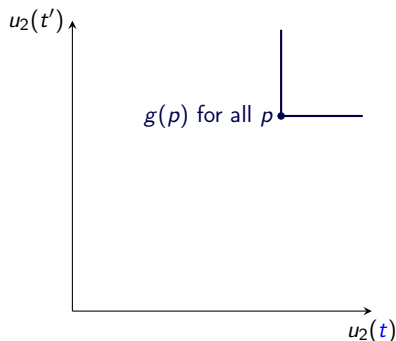
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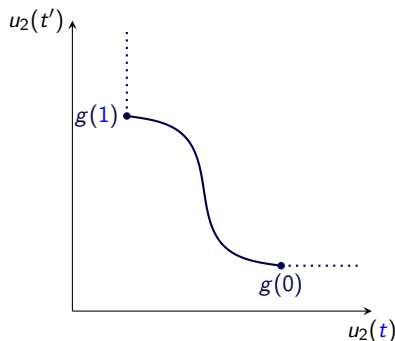
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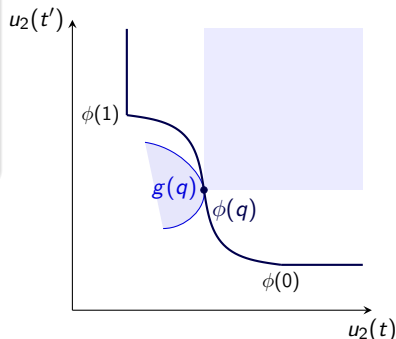
Nondominance

Assumption (Nondominance)

There is a continuous selection $\phi(q)$ of agent payoffs in $g(q)$, $q \in \Delta T_2$, st. for each $q \neq q'$ and each $u \in g(q')$, there is $t \in \text{supp } q'$ with

$$u_2(t) < \phi(t|q).$$

- ϕ tracks the frontier
- each u can be “projected” on enhanced ϕ
- the set of vectors above that are IC is called $\mathcal{R}$
 - under principal’s verifiability, the set depends on p_1 only.



Mechanism design

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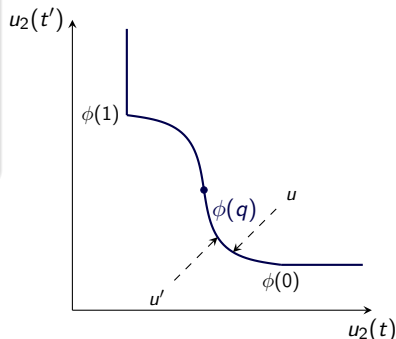
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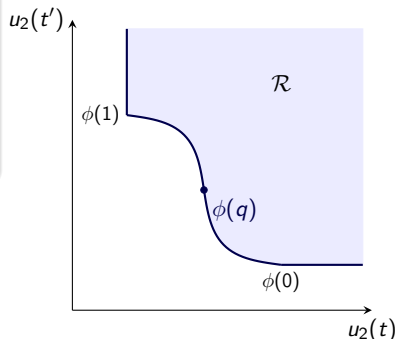
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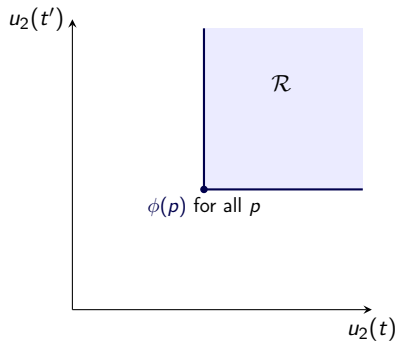
Nondominance: examples

- *Second-price auction*: belief-free

$$\phi(t_2) = \mathbb{E}_{p_1}(t_2 - t_1)^+$$

- *First-price auction*

- ▶ $\phi(t|q)$ is the best response payoff against the equilibrium bidding strategy of the opponent,
- ▶ Nondominance fails,
- ▶ all payoffs dominate $\phi(0+)$ (Maskin Riley 00),
- ▶ *Nondominance+*: g is a “limit” of games that satisfy Nondominance.



Mechanism design

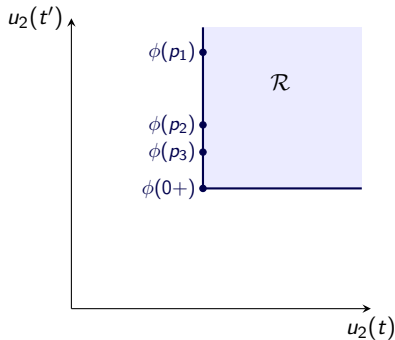
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Mechanism design

Payoff characterization under Nondominance

- Define the principal's payoff from the efficient allocation given the agent's u as

$$u_1^*(u; p) = \sup\{p_1 \cdot u'_1 : u' \in \mathcal{IC}(p), u'_2 = u\}.$$

Theorem 2

Under Nondominance, for each p ,

$$v^*(p) = \sup_{u \in \mathcal{R}} u_1^*(u; p).$$

No mediator is needed.

- IR-style constraint: u must be “above” $\bigcup_q g(q)$ (i.e., $u \geq \phi(\cdot|q)$ for some q).

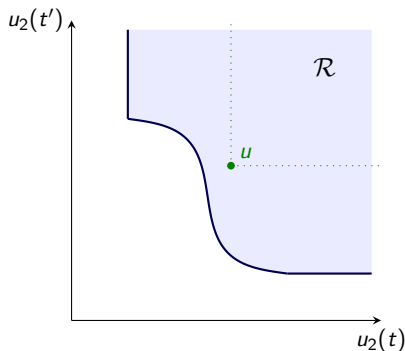
Mechanism design

Lower bound: a single allocation

- For any $u \in \text{int } \mathcal{R}$, let

$$m = (u_1^*(u), u).$$

- u is the agent's equilibrium payoff: if rejected $\implies \phi(q^R)$,
- no other equilibrium:
 - for any q and $v \in g(q)$, there is t_2 with $q(t_2) > 0$ that strictly prefers u ,
 - that includes enhanced vectors.



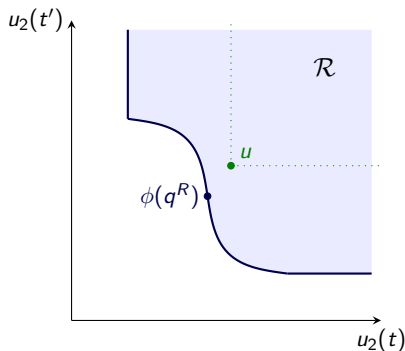
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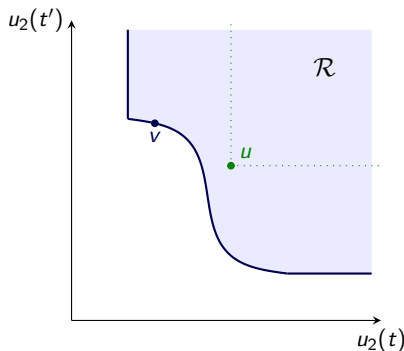
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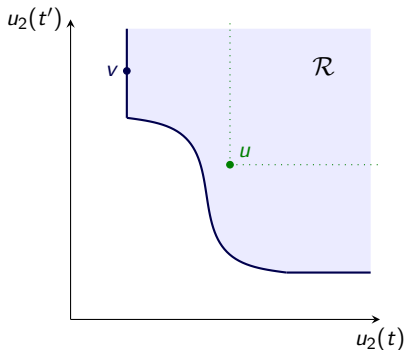
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Mechanism design

Regular g

- $L_g(u) = \{q : (u_1, u) \in g(q) \text{ for some } u_1\}$ - beliefs at which u is an equilibrium payoff of the agent.

Definition

g is regular if:

- $u.h.c.$, non-empty valued, bounded, enhanced,
- convex-valued $\Leftarrow$ public randomization,
- $L_g(u)$ convex $\Leftarrow$ agent's cheap talk.

- Bi-convexity issues (Forges 84, 90), (Aumann Hart 86, 03).
- Communication equilibria (Forges 86), (Myerson 82); Bayes correlated equilibria (Bergemann Morris 16).

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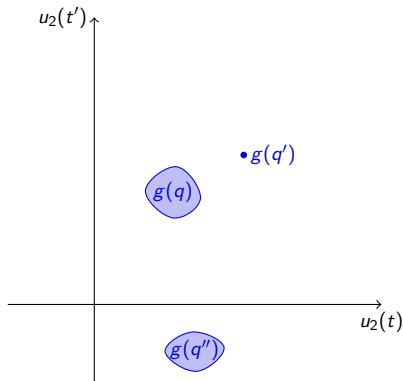
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Mechanism design

Regular payoff correspondence

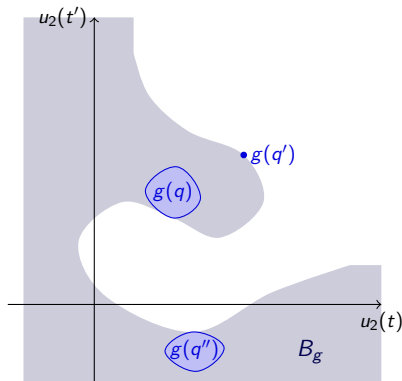
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- $B_g = \bigcup_q g(q)$ body of g
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- $\mathcal{R}$ open connected component of $\mathcal{IC}_2 \setminus B_g$ that contains $+\infty$



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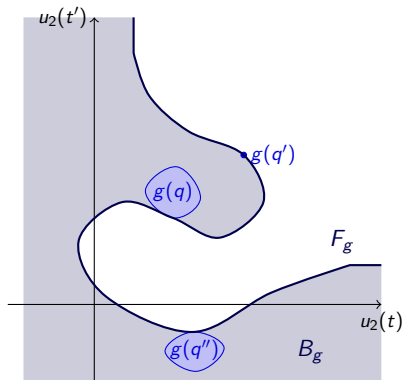
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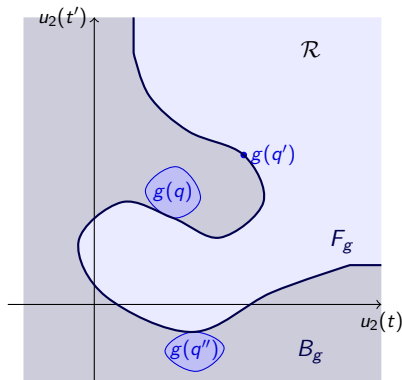
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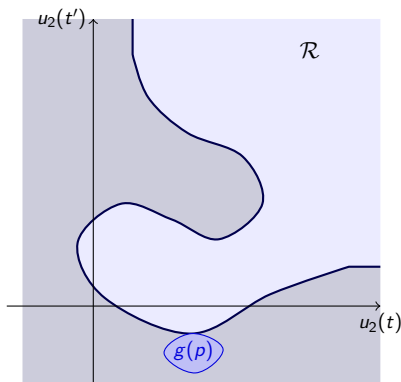
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Mechanism design

Lower bound under regularity

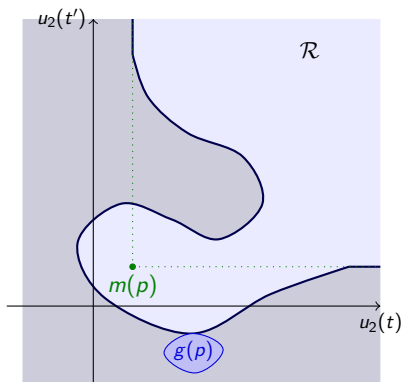
- default game g
- offer m
 - ▶ single allocation mechanism may have partial revelation equilibria
 - ▶ use mediator to avoid crossing
- possible rejection equilibria
- tolerated rejections:
 $\{u \in g(p) : u_1 \geq v_1\}$
- excluded: dominated by the rest of $g(p)$
- $\mathcal{R}(p, v_1)$ - connected component after exclusions



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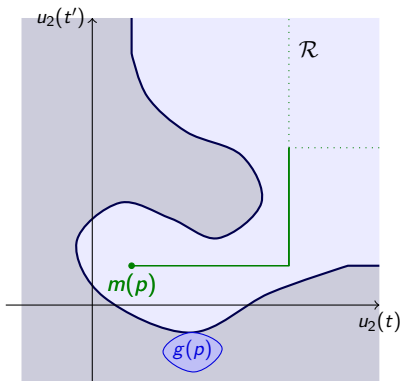
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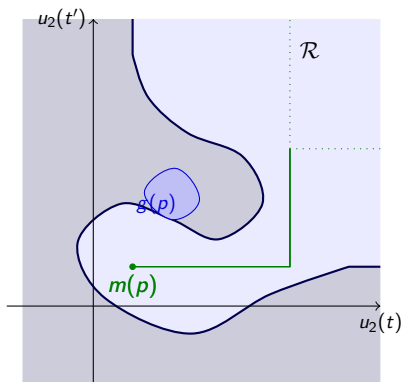
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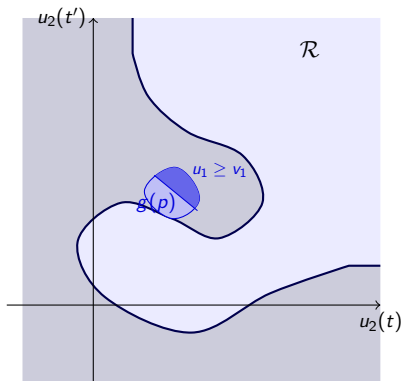
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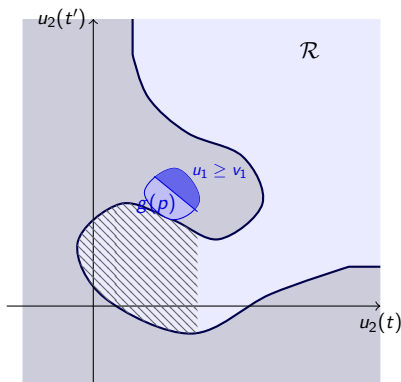
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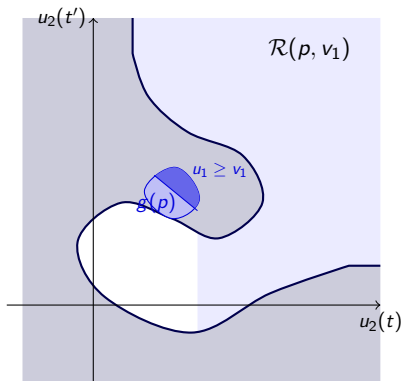
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Mechanism design

Payoff characterization under regularity

Theorem 3

Suppose that g is regular. Then

$$v^*(p) = \sup_{v_1} \min \left\{ \sup_{u \in \mathcal{R}(p, v_1)} u_1^*(u), v_1 \right\}.$$

- flexible IR constraint:
 - ▶ principal may consider rejections with payoffs above v_1 worthwhile,
- Nondominance: constraint is wlog not “flexible”
 - ▶ any continuation equilibrium can be directly replicated by the constructed mechanism.
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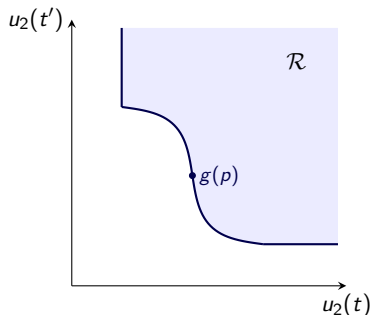
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Mechanism design

Upper bound

WTS: For each m , there is an equilibrium with payoffs in $\mathcal{R}$:

- Accept: $m(p) \in \mathcal{R}$.
- Reject: $m(q^A) \leq g(p)$.
- otherwise, a crossing $u \in m(q^A) \cap g(q^R) \cap \mathcal{R}$ with $p \in [q^R, q^A]$:
 - two types: intermediate value theorem,
 - more types: degree theory.

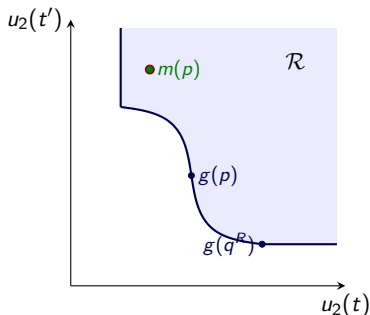


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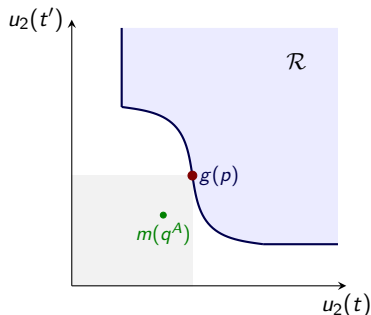


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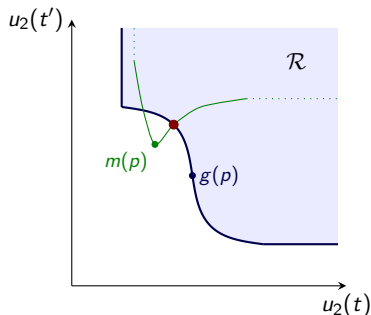


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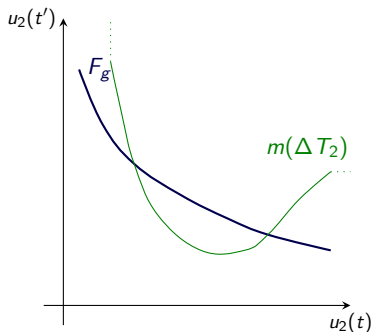
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Mechanism design

Crossing Lemma: proof

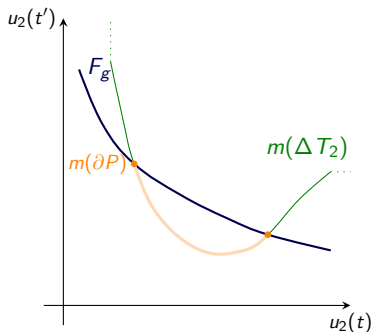
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- $P = \{q : m(q) \text{ below } F_g\}$,
- ∂P : candidate crossings
- no Accept $\implies p \in P$
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Mechanism design

Crossing Lemma: proof

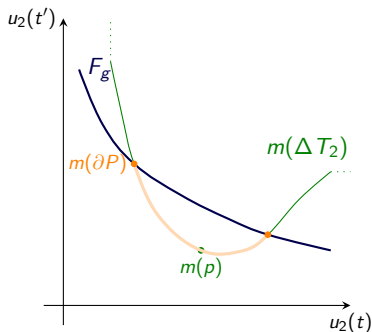
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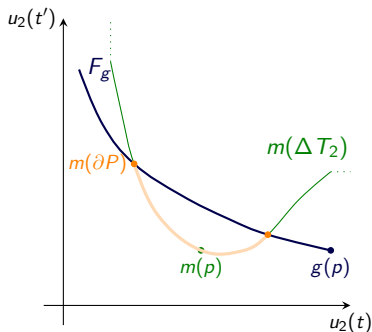
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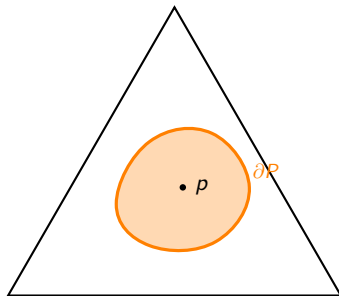
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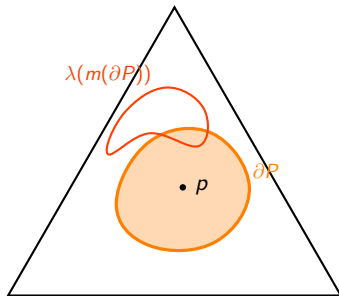
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Mechanism design

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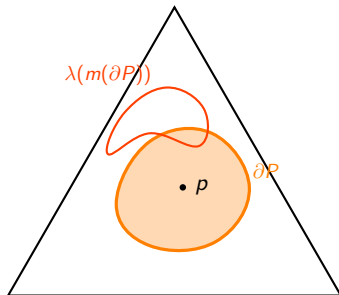
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Mechanism design

Crossing Lemma: proof

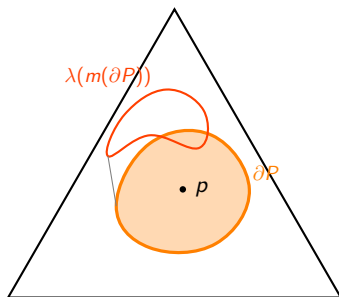
- $P \ni p$;
- $p \notin$ “inside of” $\lambda(m(\partial P))$
- regularity $\implies$ correspondence $L_g(u) = g^{-1}(u)$ is approximated by continuous λ .
- $\Psi(q, s) = (1 - s)q + s\lambda(m(q))$,
 $q \in \partial P$
- crossing: $\Psi(q, s) = p$
- topological degree theory $\implies$ crossing exists



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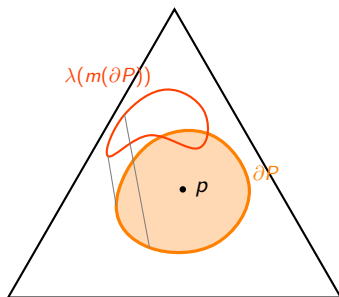
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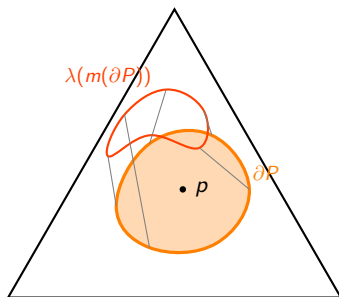
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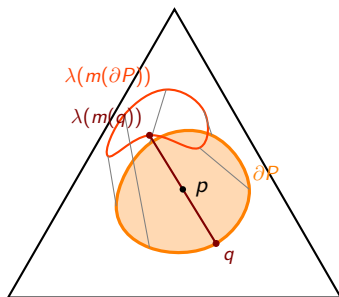
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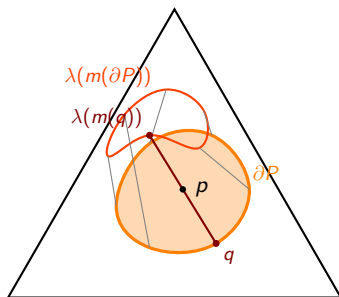
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Conclusions

- Informed principal and mechanism design problem with default continuation game g
- Assumption: principal types are verifiable!
- **Informed principal:** minimum equilibrium payoff is equal to the concavified value of the mechanism design problem
- **Mechanism design:** under Nondominance or regularity, the value is the solution to the principal's payoff maximization subject to a flexible IR constraint.